Lexical Task Heads beyond Simple Tasks

Schematic project summary

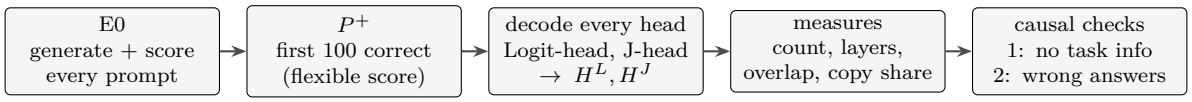
Filippo Merlo

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Marks: ✓ works / supports the hypothesis, ✗ does not, ∼ mixed or task-dependent. Full methods and tables: `proposal-minimal.tex`.

1 Setup in one page

Question	Does the lexical-task-head method of Yang et al. (2026) find task-carrying heads on the complex, rule-defined tasks of Fu et al. (2026)? Does Head-JLens (J-head) find heads the per-head logit lens (Logit-head) misses?
Model	Llama-3.1-8B-Instruct (bf16; only model with released Head-JLens matrices).
Readouts	Logit-head: head write $c_{l,h}$ decoded by the unembedding. J-head: head output $z_{l,h}$ mapped through the fitted Jacobian $J_{l,h}$, then decoded.
Simple tasks	14 Todd et al. tasks (antonym, capitals, translation, ...). Rule style: 5 instruction templates; example style: 1/5/10/20/30 shots; zero-shot prompt. Raw text, no chat template.
Complex tasks	Fu et al.'s 5 tasks, easy level: Set Game, Tapatan, Operator Function, Noun Class Agreement (NCA), Lexical Category Inference (LCI). 300 items (LCI 600). Rule style: original + 6 GPT-5.5 wordings; example; combined; input block only (no task info). Chat template, greedy, up to 1,024 tokens.
Head criterion	Head is a lexical task head on a prompt if ≥ 1 of its top-10 tokens is in the task's term list; for the style if this holds on $\geq 10\%$ of P^+ (first 100 correct prompts of the most accurate wording).
Term lists	Simple: Yang et al.'s (2–22 token ids). Complex: GPT-5.5, checked by hand (114–340 token ids).
Causal check 1	Yang App. G.1: patch the mean head output ($\times\alpha$) at the last token of prompts *without* task info; control = mean of another task at the same heads.
Causal check 2	Yang's released script: patch prompts that *give* the task but were answered wrongly (baseline 0). Complex tasks: control added.
Head sets patched	H^L, H^J, and J-only $H^J \setminus H^L$.



2 Experiments and outcomes

Experiment	Design	Outcome (short)	
E0 — behavioural baseline			
Simple tasks	14 tasks × 11 prompts, every item.	All tasks well above zero-shot; accuracy rises 1→5 shots then flat. Template spread largest on translation, next/prev-item, park–country, synonym.	✓
Complex tasks	5 tasks × 10 prompts, 300 items.	NCA, LCI answered straight away, above chance. Set Game, Op. Function, Tapatan always reason first. Set Game and Op. Function solved from rules, hardly from examples alone. Tapatan near chance everywhere. Set Game solved on 97/300 even without task info.	~
Scoring audit	Every reply saved; strict (source) vs flexible vs GPT-5.5 judge; disagreement reasons.	Much of the wording spread is scoring: Op. Function 0.01–0.99 across wordings → 1.00 everywhere (missing `\boxed`); translation spread mostly quotes/capitals; first-token score marks any number correct.	✓
E1 — simple tasks (14 tasks × rule/example × 2 readouts)			
Head sets	100 prompts per style; most accurate template / 5-shot.	Logit-head finds heads on every task. J-head finds more on 23/28 pairs; its extra heads sit in layers 10–19 (Logit-head mostly ≥20). J-head contains all of Logit-head only on 4/28. Example heads mostly inside rule heads.	✓
Check 1 (zero-shot)	Means per wording, 15 α values, control mean.	Logit-head: gain ≥10 items on capitals, landmark, park, product (+ currency, next-item, plural with examples). Nothing on antonym, synonym, present–past, prev-item, translations. Best around $\alpha = 4$. Control mostly low (not on park–country, next-item).	~
Check 1, J-head	same	J-head ≥ Logit-head on most tasks; J-only heads have effects of their own.	✓
Cross-style means	Rule heads × example means and vice versa.	Same tasks gain, often as much or more.	✓
Check 2 (1-shot wrong)	Yang's released script, no control.	Gains on all 14 tasks; reproduces Yang's Figs. 57/62 task order. J-head far stronger on next/prev-item with examples.	✓
E1b — heads from several templates (rule style, simple tasks)			
$k = 5$ (all templates)	Heads passing under all 5 templates; both checks.	Better on fact tasks (capitals, park, product) only; worse elsewhere. Beats E1 on ≤4/14 tasks.	✗
$k = 1$ (union)	Exploratory.	Helps J-head on present–past (0→16 items), synonym, prev-item, control flat. Hurts Logit-head on fact tasks.	~
$k = 3$	Exploratory.	Mostly the E1 heads; behaves like E1.	–
$k = 1$, shared	J-head union, kept only heads in the union sets of ≥ 2 tasks (keeps 15–62 of 19–74 heads per task); both checks.	**Not successful.** Removes most of the union's gains: present–past 16→1 items (check 1), 0.64→0.45 (check 2); synonym 0.21→0.13. Only country–capital rises (0.60→0.69). Shared heads do not carry the task better.	✗
$k = 1$, 4 present–past heads	The 4 heads only present–past's union set contains (L19H10, L19H24, L20H28, L21H9; none in E1); both checks, present–past only.	Check 1: no effect alone (0.01); needs the other 15 heads too. Check 2: 0.63, as the whole 19-head union (0.64); rises with α up to 10, real past forms (60/61 distinct replies); $\alpha = 0$ gives 0.02.	~
Replies, present–past	Each patched reply sorted by kind, per head set and α (J-head, rule means; notebook below).	Shared heads make the model *name* the task ("past tense of assess": 65% of replies in check 1, 73% in check 2 at $\alpha = 10$) without doing it. The 4 unshared heads make it give *a form of the input*: from a bare prompt it repeats the verb (44%), when the prompt shows the task they fix the inflection. Together (union) they give the answer. One task; heuristic reply kinds.	–
Replies, all tasks	Same, every task.	Many patched replies contain the expected answer after a prefix the flexible score rejects ("Capital – Tbilisi", "plural needles", "Country – Russia, . . . "): 19–33% of E1 J-head replies at $\alpha = 4$ on capitals, landmark, antonym, synonym, singular–plural. Not all are answers ("opposite of solid" for *hollow*).	–
Retrieval heads	Yang et al.'s retrieval heads (their code): Definition 1 with the prompt's own answer token as target, on E1's identification prompts; overlap with every lexical set, against layer-matched chance.	Mostly distinct from lexical task heads: 2–9% of each set's heads are retrieval heads, at chance under Logit-head (0.6–1.2×), slightly above under J-head (1.1–1.7×, highest for $k = 5$). None on the translation tasks. Of the 4 present–past unshared heads only L21H9 is one (15% of prompts).	–

3 Hypotheses

H1 replication	Holds with Yang's released setting (check 2); fails with the procedure their paper describes (check 1: no gain on 7 tasks).	∼
H2 complex < simple	Causal part agrees (no task-specific effect on complex tasks); head-count part untestable (term-list length).	∼
H3 J-head ⊇ Logit-head	Fails: J-head misses part of the Logit-head set on 24/28 simple and all complex pairs. J-only heads causal on simple tasks, not on complex.	✗
H4 all-template heads	Falsified: better on ≤4/14 tasks.	✗

4 Design variants tried

Date	Tried first	Replaced by	Why / effect
10-27	Rules as main style, examples secondary	Rule and example treated alike	No main condition (user decision).
10-27	Drop prompts not answered straight away	Reasoning mode for Set Game, Op. Function, Tapatan	They are never answered straight away.
10-27	Wordings/terms via ChatGPT	GPT-5.5 via Azure API, checked by hand	API instead of the chat interface.
10-27	All difficulty levels	Easy level only	User decision.
10-27	Reasoning mode: 10 items, $\alpha \in \{0, 1, 4, 10\}$	100 items, 11 α values (10-30)	Same 100 items as the other tasks.
10-28	Prompts and means pooled over wordings	One wording for heads, one mean per wording (Yang's code)	Matches the source exactly.
10-28	Yang's batching without attention mask	Padding masked	Model attended to padding.
10-28	Check 1 only (zero-shot)	Check 1 + check 2 (released script)	Only check 2 reproduces Yang's figures.
10-29	6,000-token replies, stop at the answer; direct vs reasoning split	1,024 tokens for all five complex tasks, one scheme	4/9,000 replies correct after token 1,024.
10-29	Heads also read at a pre-answer position	Last prompt token only	Dropped with its results.
10-30	Tapatan scored by its first line	First label anywhere	First line is the label in 1/3,000 replies.
10-30	First-token (simple) / strict (complex) score for heads and checks	Flexible score; E1, E2 rerun	Strict scoring created spurious wording differences.
10-31	Complex check 2 without control	Control mean added	Zero-ablation and label shifts alone gave gains.
11-02	E1b with $k = 5$ only	$k \in \{1, 3, 5\}$	After $k = 5$ failed; exploratory.

Inspecting replies: `notebooks/patching_outputs.py` (marimo) shows the simple-task patched replies for every manipulation: task, check, procedure (E1, $k = 1/3/5$, shared/unshared), readout, head and mean style, mean source and wording, α; accuracy, the share of the most common reply, reply kinds, and each item's replies.

Not pursued: function vectors, function-specific heads, hidden-state differences (scope, 10-27); fp32 Head-JLens fit (≈136 GPU-h, declined); hand-labelled validation of the judge.

5 What seems to work, what does not

- ✓ **Scoring matters first.** Flexible scoring removes most of the apparent wording sensitivity; always check it before reading wording effects.
- ✓ **Released-code setting (check 2)** is what reproduces Yang et al.; the paper's zero-shot check works only on fact-like tasks.
- ✓ **J-head on simple tasks:** more heads, earlier layers (10–19), and the added heads carry the task causally. It does not contain the Logit-head set.
- ∼ **Which heads to keep** depends on the task type: all-template heads for fact tasks, union of templates for morphology-like tasks under J-head.
- ✗ **Complex tasks:** heads are found, but no causal check shows a task-specific effect. Prompts without task

info elicit no attempt; on wrong-answer prompts, any perturbation (zero, control, label shift) helps as much.

- ✗ **Head counts on complex tasks** are not interpretable with long term lists and tiny P^+ sets.
- **Controls are essential** on long-reasoning and Yes/No tasks: without them, check 2 looked positive.
- ∼ **Retrieval heads** (heads that already write the answer token) are few among lexical task heads, but some effects rest on them: landmark–country's on its 10, present–past's union gain on L21H9. Elsewhere removing them changes little or helps (synonym, product–company, prev-item under J-head). Entity-answer tasks may be carried by retrieval more than by task heads.
- ∼ **Two roles on present–past** (one task): heads shared across tasks make the model *name* the task; L21H9 makes it give *a form of the input*. Either alone fails from a bare prompt; with an example in the prompt, L21H9 alone suffices.
- **Read the replies, not only accuracy:** 19–33% of patched replies on several tasks contain the answer after a prefix the flexible score rejects, and part of a "loss" can be a format change (landmark–country under J-head). `notebooks/patching_outputs.py` shows them.